Batch 5 · Take-home assignment · Version 2

A research assistant that remembers

Build a multi-agent assistant that helps employees find trustworthy answers in company documentation and continue useful work across conversations.

Submit: Git repository and accessible LangSmith project.
Deadline: Oct 27th (Wed) EOD.

The problem

Asterline Labs is a fictional workflow-monitoring company. Its documentation includes product specifications, pricing, policies, release notes, and incident reports. An employee needs to ask questions, investigate issues, and return later without having to repeat all relevant context.

The assistant should answer from the supplied documents, recognise uncertainty, and remember selectively. Documents may disagree or describe different points in time. Users may correct earlier information or ask the assistant to forget it.

What you receive

`corpus.jsonl` contains 75 chunks from 25 documents, including source identifiers and metadata. Use this as the knowledge base for company facts. Keep it unchanged. No prior knowledge of Asterline is needed.

Expected outcomes

- **Trustworthy research:** Answer questions using relevant documentary evidence, cite the supporting chunk IDs and titles, and handle conflicting or missing information honestly.
- **Useful collaboration:** Use at least three agents with distinct responsibilities. Their collaboration should contribute to the result and be understandable from the execution trace.
- **Continuity:** Support follow-up questions and retain useful context across separate sessions, including after restarting the application.
- **Thoughtful memory:** Design what the assistant remembers, what it avoids storing, and how it uses that information. Support corrections and requests to forget. One user's private context must not be exposed to another user.
- **Usability:** Provide a simple interface for conversations and a way for users to inspect and manage what is remembered about them.

You choose the architecture, memory model, orchestration, storage, and evaluation approach. We are interested in the reasoning behind those choices, not a particular framework or number of components.

Build, demonstrate, and explain

Constraints

- Use local embeddings and a local vector store. Choose an accessible generation model and document any costs or account requirements; paid services are not required.
- Use LangSmith to make agent behaviour and important decisions inspectable. Use synthetic user data and keep secrets out of code, stored memory, and shared traces.
- Keep remembered context distinct from authoritative company evidence. A remembered statement or previous answer is not sufficient proof of a company fact.
- Provide a documented setup and one command to launch the application. Handle provider failures and rate limits reasonably.

A local demonstration is sufficient. Production authentication, deployment, and visual polish are not required. A user selector may simulate separate users, but their stored context must remain isolated.

What to submit

1. **A working repository:** Application code, unchanged corpus, setup instructions, dependencies, and an environment-variable example without secrets.
2. **A short design note:** Explain the agent responsibilities, information flow, memory approach, and important trade-offs. Include an architecture diagram and identify limitations.
3. **Evidence that it works:** Create your own reproducible evaluation covering research quality and memory behaviour. Explain what you measured and why, include results and representative failures, and show one improvement driven by those findings.
4. **Inspectable runs:** Provide LangSmith access and links to representative runs that demonstrate both successful behaviour and a difficult or failed case. Grant access to `radialpulse@nxtwave.co.in` or provide an accessible project link.
5. **A brief AI-use disclosure:** State which tools you used, what they helped produce, and how you checked their output. If you used none, say so.

How your work will be assessed

Area	Weight
Memory design and dependable behaviour	25%
Grounded research and answer quality	20%
Meaningful multi-agent collaboration	15%
Evaluation, failure analysis, and improvement	20%
Observability	10%
Code quality and reproducibility	10%

Walkthrough and ownership

AI coding tools are allowed. You remain responsible for understanding and validating the submitted system. Be prepared for a short live walkthrough: explain a design decision, follow a question through a trace, investigate a failure, and adapt the system to a small new requirement within the assignment's scope.

We may try questions and conversation sequences that are not in your own tests. A polished demonstration alone is not sufficient; clear reasoning, reproducible behaviour, and an honest account of limitations matter.

Package contents and README

This PDF contains the complete assignment brief, README, and all 75 dataset records, including every metadata field. The three original files are embedded as PDF attachments, so the unchanged corpus.jsonl can be extracted with a PDF reader that supports attachments.

Original README.txt

BATCH 5 — ASSIGNMENT

The candidate package contains only this README, the assignment brief, and corpus.jsonl.

Corpus: 25 documents / 75 chunks.

Deadline: Oct 27th (Wed) EOD.

The architecture and evaluation approach are yours to design.

Using the dataset

The following pages show every corpus record in its original order. Field names and values are preserved. Use the embedded original corpus.jsonl file as the application input; the formatted pages are provided for reading.

The deadline wording is reproduced exactly as supplied in the original package.

01. Company overview - company:0

chunk_id: company:0
doc_id: company
title: Company overview
category: onboarding
owner: Asterline Documentation
source_url: kb://asterline/onboarding/company
published: 20260301
version: 1

Asterline Labs is a fictional workflow-monitoring software company. Its product, Asterline, collects workflow runs, helps teams investigate failures, and sends operational alerts. The company has offices in Pune and Dublin. This knowledge base is the only approved evidence source for this exercise.

02. Company overview - company:1

chunk_id: company:1
doc_id: company
title: Company overview
category: onboarding
owner: Asterline Documentation
source_url: kb://asterline/onboarding/company
published: 20260301
version: 1

A workspace is a billing and access boundary. Projects belong to a workspace and each project has its own ingestion key. A workflow run contains a run_id, workflow_name, started_at, status, and optional attributes. A retry is a separate attempt associated with the same run_id.

03. Company overview - company:2

chunk_id: company:2
doc_id: company
title: Company overview
category: onboarding
owner: Asterline Documentation
source_url: kb://asterline/onboarding/company
published: 20260301
version: 1

This overview is introductory, not a source for commercial limits or incident procedures. Use the dated pricing, product, policy, and operational documents for those questions. An employee or customer may describe their own workspace context; that statement does not establish a product-wide capability.

04. Plans and billing - pricing:0

chunk_id: pricing:0
doc_id: pricing
title: Plans and billing
category: pricing
owner: Asterline Documentation
source_url: kb://asterline/pricing/pricing
published: 20260310
version: 3

Launch costs $59 per month and includes 4 seats and 1 million run records per month. Team costs $449 per month and includes 20 seats and 15 million run records per month. Enterprise pricing is negotiated; no list price is published. Annual billing gives a 12% discount on Launch and Team base subscriptions.

05. Plans and billing - pricing:1

chunk_id: pricing:1
doc_id: pricing
title: Plans and billing
category: pricing
owner: Asterline Documentation
source_url: kb://asterline/pricing/pricing
published: 20260310
version: 3

Launch retains searchable run records for 21 days; Team for 180 days; Enterprise for 540 days. Run-record overage is $1.20 per 10,000 records on Launch and $0.70 per 10,000 on Team. Enterprise overage is negotiated. Extra Team seats cost $16 per seat per month; Launch cannot buy extra seats.

06. Plans and billing - pricing:2

chunk_id: pricing:2
doc_id: pricing
title: Plans and billing
category: pricing
owner: Asterline Documentation
source_url: kb://asterline/pricing/pricing
published: 20260310
version: 3

Launch allows 6 alert rules per project, Team 75, and Enterprise 600. Warehouse Export is unavailable on Launch and included on Team and Enterprise. SAML SSO is Enterprise only. Team and Enterprise can create EU projects. Launch supports US projects only. A trial lasts 10 days with Team features and a total cap of 100,000 run records.

07. Service and support terms - sla:0

chunk_id: sla:0
doc_id: sla
title: Service and support terms
category: pricing
owner: Asterline Documentation
source_url: kb://asterline/pricing/sla
published: 20260310
version: 2

Only Enterprise has a contractual monthly uptime SLA, set at 99.92%. Service credits are 10% below 99.92%, 25% below 99.5%, and 50% below 99.0%; the highest applicable tier applies, and credits are not added together.

08. Service and support terms - sla:1

chunk_id: sla:1
doc_id: sla
title: Service and support terms
category: pricing
owner: Asterline Documentation
source_url: kb://asterline/pricing/sla
published: 20260310
version: 2

Launch receives email support within 2 business days. Team receives email and chat support within 6 business hours. Enterprise receives 24-hour support with a 45-minute response target for P1 cases. Response targets are not resolution promises.

09. Service and support terms - sla:2

chunk_id: sla:2
doc_id: sla
title: Service and support terms
category: pricing
owner: Asterline Documentation
source_url: kb://asterline/pricing/sla
published: 20260310
version: 2

Planned maintenance announced at least 72 hours ahead is excluded from the uptime calculation. Customers must request service credits within 20 calendar days after the affected month ends. Trial workspaces do not have a contractual uptime SLA.

10. Run ingestion API - ingest:0

chunk_id: ingest:0
doc_id: ingest
title: Run ingestion API
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/ingest
published: 20260310
version: 2.3

POST /v2/runs accepts at most 400 run records or 1 MB per request, whichever limit is reached first. A valid batch returns HTTP 202. Oversized requests return 413. Authentication uses a project ingestion key in the Authorization header; never paste actual keys into support conversations.

11. Run ingestion API - ingest:1

chunk_id: ingest:1
doc_id: ingest
title: Run ingestion API
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/ingest
published: 20260310
version: 2.3

The per-project ingestion rate limit is 300 requests per second on Launch, 1500 on Team, and 6000 on Enterprise. A rate-limited request returns 429 and a Retry-After header. Clients should honor the header, use bounded retries, and retain stable event identifiers when retrying.

12. Run ingestion API - ingest:2

chunk_id: ingest:2
doc_id: ingest
title: Run ingestion API
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/ingest
published: 20260310
version: 2.3

Duplicate submissions with the same event_id are suppressed for 48 hours. Records older than 96 hours are accepted with late=true. Each record supports at most 80 custom attributes and each attribute value is limited to 2 KB. Accepted records normally become searchable within 45 seconds; this is not a contractual guarantee.

13. Signals alert rules - alerts:0

chunk_id: alerts:0
doc_id: alerts
title: Signals alert rules
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/alerts
published: 20260310
version: 2.3

Signals evaluate workflow metrics every 4 minutes. Supported modes are static threshold, percentage change against a previous period, and seasonal anomaly. Seasonal anomaly requires 21 complete days of history. Rules with insufficient history remain in a warming-up state.

14. Signals alert rules - alerts:1

chunk_id: alerts:1
doc_id: alerts
title: Signals alert rules
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/alerts
published: 20260310
version: 2.3

Supported destinations are email, Slack, webhook, and PagerDuty. PagerDuty was added in version 2.3. Limits are 6 rules per project on Launch, 75 on Team, and 600 on Enterprise. A muted rule still evaluates but does not deliver notifications.

15. Signals alert rules - alerts:2

chunk_id: alerts:2
doc_id: alerts
title: Signals alert rules
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/alerts
published: 20260310
version: 2.3

Since patch 2.3.1 the scheduler uses a monotonic clock and replays missed evaluation windows after recovery. This fixes the clock-adjustment defect in INC-2026-03. Replayed evaluations are tagged recovered=true so operators can distinguish delayed evaluations from live ones.

16. Warehouse Export specification - export:0

chunk_id: export:0
doc_id: export
title: Warehouse Export specification
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/export
published: 20260205
version: 2.2

Warehouse Export sends selected run records to a customer warehouse. Team supports BigQuery and Snowflake. Enterprise supports BigQuery, Snowflake, and Redshift. Launch has no Warehouse Export. Redshift support began in version 2.2.

17. Warehouse Export specification - export:1

chunk_id: export:1
doc_id: export
title: Warehouse Export specification
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/export
published: 20260205
version: 2.2

Exports can run as frequently as every 20 minutes. The destination receives an export_job_id and a record_hash. Since patch 2.1.2 the pair (export_job_id, record_hash) is the idempotency key; this replaced position-based deduplication after INC-2025-12.

18. Warehouse Export specification - export:2

chunk_id: export:2
doc_id: export
title: Warehouse Export specification
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/export
published: 20260205
version: 2.2

Exports use a customer-managed warehouse credential stored in the secret vault. A failed job retries up to three times with backoff and then enters failed status. Operators can re-run failed jobs after resolving permissions. Deleting an Asterline record does not automatically delete copies already exported into the customer warehouse.

19. Run Explorer specification - traces:0

chunk_id: traces:0
doc_id: traces
title: Run Explorer specification
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/traces
published: 20260205
version: 2.2

Run Explorer shows the ordered attempts and status transitions for a workflow run. It is an operational timeline, not a recording of a user screen. Explorer is included on Team and Enterprise and is unavailable on Launch.

20. Run Explorer specification - traces:1

chunk_id: traces:1
doc_id: traces
title: Run Explorer specification
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/traces
published: 20260205
version: 2.2

Explorer timeline retention is 14 days on Team and 60 days on Enterprise. This is separate from searchable run-record retention. Sampling is 50% on Team and 100% on Enterprise, selected deterministically by run_id.

21. Run Explorer specification - traces:2

chunk_id: traces:2
doc_id: traces
title: Run Explorer specification
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/traces
published: 20260205
version: 2.2

Users can filter timelines by workflow name, status, environment, and custom attributes. A shareable timeline link requires workspace access and does not grant anonymous viewing. Deleting a project removes its timeline access immediately.

22. AQL query reference - query:0

chunk_id: query:0
doc_id: query
title: AQL query reference
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/query
published: 20260310
version: 2.3

AQL is a read-only query language. Example: RUNS WHERE status = failed SINCE 7d COUNT BY workflow_name. Queries return at most 8000 rows and time out after 50 seconds in version 2.3. Earlier versions used a 25-second timeout.

23. AQL query reference - query:1

chunk_id: query:1
doc_id: query
title: AQL query reference
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/query
published: 20260310
version: 2.3

Per-workspace query concurrency is 6 on Launch, 30 on Team, and 120 on Enterprise. Queries beyond this limit are rejected with 429. Ingestion limits are per project and must not be confused with these workspace query limits.

24. AQL query reference - query:2

chunk_id: query:2
doc_id: query
title: AQL query reference
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/query
published: 20260310
version: 2.3

Queries read the hot store for the most recent 5 days and object storage for older data. Results are cached for 8 minutes. An explicit bypass_cache option is available to workspace admins; it does not increase query limits or retention.

25. SDK integration guide - sdk:0

chunk_id: sdk:0
doc_id: sdk
title: SDK integration guide
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/sdk
published: 20260205
version: 2.2

Official SDKs are available for JavaScript, Python, Java, and Go. The JavaScript SDK supports Node.js services. SDKs submit workflow lifecycle events and are not browser session-recording libraries.

26. SDK integration guide - sdk:1

chunk_id: sdk:1
doc_id: sdk
title: SDK integration guide
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/sdk
published: 20260205
version: 2.2

SDKs flush a batch every 15 seconds or when it reaches 100 records. On retry they retain the original event_id. Buffer capacity is 5000 records per process by default; when full, the oldest unsent record is dropped and a warning is logged.

27. SDK integration guide - sdk:2

chunk_id: sdk:2
doc_id: sdk
title: SDK integration guide
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/sdk
published: 20260205
version: 2.2

Version 2.2 introduced an async Python client. Applications must call flush during graceful shutdown and allow up to 10 seconds for completion. Use separate project keys for development and production. Keys are configured through environment variables.

28. Regions and workspace setup - regions:0

chunk_id: regions:0
doc_id: regions
title: Regions and workspace setup
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/regions
published: 20260115
version: 2.1

Asterline has a US region in Virginia and an EU region in Dublin. US is the default. EU projects are available on Team and Enterprise starting with version 2.1. Launch workspaces can create only US projects.

29. Regions and workspace setup - regions:1

chunk_id: regions:1
doc_id: regions
title: Regions and workspace setup
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/regions
published: 20260115
version: 2.1

Region is selected when a project is created and cannot be changed in place. To move future ingestion, create a project in the target region and update SDK credentials and endpoints. Existing historical records are not transferred by this procedure.

30. Regions and workspace setup - regions:2

chunk_id: regions:2
doc_id: regions
title: Regions and workspace setup
category: product
owner: Asterline Documentation
source_url: kb://asterline/product/regions
published: 20260115
version: 2.1

A workspace may contain projects in both supported regions if its plan permits EU projects. Queries and alert rules operate within a project region. No launch date has been announced for an Asia-Pacific region.

31. Version 2.0 release notes - release-20:0

chunk_id: release-20:0
doc_id: release-20
title: Version 2.0 release notes
category: release-notes
owner: Asterline Documentation
source_url: kb://asterline/release-notes/release-20
published: 20250812
version: 2.0

Version 2.0 shipped on 2025-08-12. It introduced the Queuefin ingestion pipeline and Run Explorer for Team and Enterprise. Launch ingestion was limited to 180 requests per second per project at this release.

32. Version 2.0 release notes - release-20:1

chunk_id: release-20:1
doc_id: release-20
title: Version 2.0 release notes
category: release-notes
owner: Asterline Documentation
source_url: kb://asterline/release-notes/release-20
published: 20250812
version: 2.0

Patch 2.0.1 shipped on 2025-08-22 and fixed the consumer configuration mismatch responsible for INC-2025-08. The rollout now checks consumer heartbeat configuration before advancing to production.

33. Version 2.0 release notes - release-20:2

chunk_id: release-20:2
doc_id: release-20
title: Version 2.0 release notes
category: release-notes
owner: Asterline Documentation
source_url: kb://asterline/release-notes/release-20
published: 20250812
version: 2.0

Version 2.0 retained a 25-second AQL query timeout. It supported email, Slack, and webhook alert destinations. EU projects, Redshift export, and PagerDuty were not introduced in this release.

34. Version 2.1 release notes - release-21:0

chunk_id: release-21:0
doc_id: release-21
title: Version 2.1 release notes
category: release-notes
owner: Asterline Documentation
source_url: kb://asterline/release-notes/release-21
published: 20251208
version: 2.1.2

Version 2.1 shipped on 2025-11-18. It introduced EU projects in Dublin for Team and Enterprise and seasonal anomaly alerting. Launch ingestion increased from 180 to 300 requests per second per project. Other plan ingestion limits stayed unchanged.

35. Version 2.1 release notes - release-21:1

chunk_id: release-21:1
doc_id: release-21
title: Version 2.1 release notes
category: release-notes
owner: Asterline Documentation
source_url: kb://asterline/release-notes/release-21
published: 20251208
version: 2.1.2

Version 2.1 replaced the query engine with Heron. The new engine changed export row ordering, exposing a flaw in position-based deduplication. INC-2025-12 documents the resulting duplicate warehouse rows.

36. Version 2.1 release notes - release-21:2

chunk_id: release-21:2
doc_id: release-21
title: Version 2.1 release notes
category: release-notes
owner: Asterline Documentation
source_url: kb://asterline/release-notes/release-21
published: 20251208
version: 2.1.2

Patch 2.1.2 shipped on 2025-12-08. It changed Warehouse Export deduplication to (export_job_id, record_hash), resolving INC-2025-12. A cleanup script was provided to affected customers; merely upgrading did not remove existing duplicate rows.

37. Version 2.2 release notes - release-22:0

chunk_id: release-22:0
doc_id: release-22
title: Version 2.2 release notes
category: release-notes
owner: Asterline Documentation
source_url: kb://asterline/release-notes/release-22
published: 20260205
version: 2.2

Version 2.2 shipped on 2026-02-05. It added Redshift Warehouse Export for Enterprise and the async Python SDK. Team warehouse destinations remained BigQuery and Snowflake.

38. Version 2.2 release notes - release-22:1

chunk_id: release-22:1
doc_id: release-22
title: Version 2.2 release notes
category: release-notes
owner: Asterline Documentation
source_url: kb://asterline/release-notes/release-22
published: 20260205
version: 2.2

The release added environment filtering in Run Explorer. It did not change Explorer sampling or retention. Enterprise timelines remained available for 60 days and Team timelines for 14 days.

39. Version 2.2 release notes - release-22:2

chunk_id: release-22:2
doc_id: release-22
title: Version 2.2 release notes
category: release-notes
owner: Asterline Documentation
source_url: kb://asterline/release-notes/release-22
published: 20260205
version: 2.2

Version 2.2 retained the 25-second AQL timeout. It did not add an Asia-Pacific region or alter the restriction against changing a project region in place.

40. Version 2.3 release notes - release-23:0

chunk_id: release-23:0
doc_id: release-23
title: Version 2.3 release notes
category: release-notes
owner: Asterline Documentation
source_url: kb://asterline/release-notes/release-23
published: 20260310
version: 2.3.1

Version 2.3 shipped on 2026-03-03. It added PagerDuty as a Signals destination and increased the AQL timeout from 25 to 50 seconds. Query row limits and concurrency limits were unchanged.

41. Version 2.3 release notes - release-23:1

chunk_id: release-23:1
doc_id: release-23
title: Version 2.3 release notes
category: release-notes
owner: Asterline Documentation
source_url: kb://asterline/release-notes/release-23
published: 20260310
version: 2.3.1

Patch 2.3.1 shipped on 2026-03-10. It fixed the clock-adjustment defect in INC-2026-03 using a monotonic scheduler clock and replay of missed evaluation windows. No alerts were lost during the incident.

42. Version 2.3 release notes - release-23:2

chunk_id: release-23:2
doc_id: release-23
title: Version 2.3 release notes
category: release-notes
owner: Asterline Documentation
source_url: kb://asterline/release-notes/release-23
published: 20260310
version: 2.3.1

Customers using the managed service receive scheduler fixes automatically. SDK upgrades are not required to receive the 2.3.1 scheduler fix. This release did not change plan prices, run-record retention, or support response targets.

43. Ingestion and query architecture - architecture:0

chunk_id: architecture:0
doc_id: architecture
title: Ingestion and query architecture
category: engineering
owner: Asterline Documentation
source_url: kb://asterline/engineering/architecture
published: 20260120
version: 2

The ingestion path is API gateway, Go ingestion workers, Kafka topic runs.raw, enrichers, Kafka topic runs.enriched, and storage writers. Both Kafka topics have 72 partitions. Raw payloads are validated before entering runs.raw.

44. Ingestion and query architecture - architecture:1

chunk_id: architecture:1
doc_id: architecture
title: Ingestion and query architecture
category: engineering
owner: Asterline Documentation
source_url: kb://asterline/engineering/architecture
published: 20260120
version: 2

Writers store recent data in ClickHouse and longer-lived records as Parquet in object storage. The hot-store window is 5 days. The Heron query engine routes reads based on the requested time range and merges results when a query crosses that boundary.

45. Ingestion and query architecture - architecture:2

chunk_id: architecture:2
doc_id: architecture
title: Ingestion and query architecture
category: engineering
owner: Asterline Documentation
source_url: kb://asterline/engineering/architecture
published: 20260120
version: 2

Query results have an 8-minute cache TTL. Kafka offsets track ingestion progress and are independent of client event_id deduplication. A successful HTTP 202 acknowledges acceptance, not completion of all downstream writes.

46. Deployment runbook - deploy:0

chunk_id: deploy:0
doc_id: deploy
title: Deployment runbook
category: engineering
owner: Asterline Documentation
source_url: kb://asterline/engineering/deploy
published: 20260120
version: 2

Production deployments use blue/green releases with canary stages at 10%, 40%, and 100%. Each stage has a 15-minute observation period. Operators record the release version and deployment identifier before starting.

47. Deployment runbook - deploy:1

chunk_id: deploy:1
doc_id: deploy
title: Deployment runbook
category: engineering
owner: Asterline Documentation
source_url: kb://asterline/engineering/deploy
published: 20260120
version: 2

Roll back automatically if the error rate exceeds 0.8% or p95 latency exceeds 900 milliseconds during a canary stage. The deployment controller also validates Kafka consumer heartbeat settings after INC-2025-08.

48. Deployment runbook - deploy:2

chunk_id: deploy:2
doc_id: deploy
title: Deployment runbook
category: engineering
owner: Asterline Documentation
source_url: kb://asterline/engineering/deploy
published: 20260120
version: 2

A rollback restores application binaries and configuration, but does not remove warehouse rows already exported. Data cleanup requires the incident-specific procedure. The incident commander decides when to resume a halted rollout.

49. On-call runbook - oncall:0

chunk_id: oncall:0
doc_id: oncall
title: On-call runbook
category: engineering
owner: Asterline Documentation
source_url: kb://asterline/engineering/oncall
published: 20250410
version: 1

The primary and secondary on-call engineers rotate weekly. This runbook lists an on-call stipend of $140 per week. The engineering manager maintains the rota and publishes handover notes on Fridays.

50. On-call runbook - oncall:1

chunk_id: oncall:1
doc_id: oncall
title: On-call runbook
category: engineering
owner: Asterline Documentation
source_url: kb://asterline/engineering/oncall
published: 20250410
version: 1

SEV-1 is a customer-facing outage: page immediately and publish a status update within 10 minutes. SEV-2 is degraded service: publish an update within 45 minutes. SEV-3 is a minor issue: update on the next business day.

51. On-call runbook - oncall:2

chunk_id: oncall:2
doc_id: oncall
title: On-call runbook
category: engineering
owner: Asterline Documentation
source_url: kb://asterline/engineering/oncall
published: 20250410
version: 1

The primary engineer becomes incident commander until relieved. The secondary maintains a timeline and gathers evidence. A post-mortem must record impact, cause, recovery, and follow-up actions without exposing customer secrets.

52. INC-2025-08 ingestion outage - incident-08:0

chunk_id: incident-08:0
doc_id: incident-08
title: INC-2025-08 ingestion outage
category: incident
owner: Asterline Documentation
source_url: kb://asterline/incident/incident-08
published: 20250825
version: 1

INC-2025-08 occurred on 2025-08-20 from 08:10 to 10:40 UTC, lasting 2 hours 30 minutes. Approximately 32% of US ingestion requests returned 503. The incident was SEV-1.

53. INC-2025-08 ingestion outage - incident-08:1

chunk_id: incident-08:1
doc_id: incident-08
title: INC-2025-08 ingestion outage
category: incident
owner: Asterline Documentation
source_url: kb://asterline/incident/incident-08
published: 20250825
version: 1

The cause was a consumer heartbeat configuration mismatch introduced during the Queuefin rollout. Rebalances repeatedly interrupted ingestion workers. SDK clients that retried recovered; clients without retry lost approximately 1.4 million records.

54. INC-2025-08 ingestion outage - incident-08:2

chunk_id: incident-08:2
doc_id: incident-08
title: INC-2025-08 ingestion outage
category: incident
owner: Asterline Documentation
source_url: kb://asterline/incident/incident-08
published: 20250825
version: 1

Patch 2.0.1 fixed the mismatch. Deployment validation now checks heartbeat settings before a canary advances. This incident concerned ingestion availability, not Warehouse Export duplication.

55. INC-2025-12 duplicate exports - incident-12:0

chunk_id: incident-12:0
doc_id: incident-12
title: INC-2025-12 duplicate exports
category: incident
owner: Asterline Documentation
source_url: kb://asterline/incident/incident-12
published: 20251210
version: 1

INC-2025-12 affected 29 Team and Enterprise customers using BigQuery or Snowflake between 2025-12-04 and 2025-12-07. Warehouse Export wrote duplicate rows. It was a SEV-2 incident.

56. INC-2025-12 duplicate exports - incident-12:1

chunk_id: incident-12:1
doc_id: incident-12
title: INC-2025-12 duplicate exports
category: incident
owner: Asterline Documentation
source_url: kb://asterline/incident/incident-12
published: 20251210
version: 1

Heron changed query result ordering, which broke position-based export deduplication. The fix replaced row position with the stable pair (export_job_id, record_hash). The fix shipped in patch 2.1.2.

57. INC-2025-12 duplicate exports - incident-12:2

chunk_id: incident-12:2
doc_id: incident-12
title: INC-2025-12 duplicate exports
category: incident
owner: Asterline Documentation
source_url: kb://asterline/incident/incident-12
published: 20251210
version: 1

Affected customers received a cleanup script for pre-existing duplicates. The fix prevented new duplicates but did not itself delete old ones. Redshift customers were not affected because that destination had not yet launched.

58. INC-2026-03 delayed alerts - incident-03:0

chunk_id: incident-03:0
doc_id: incident-03
title: INC-2026-03 delayed alerts
category: incident
owner: Asterline Documentation
source_url: kb://asterline/incident/incident-03
published: 20260311
version: 1

INC-2026-03 occurred on 2026-03-07 from 16:05 to 16:39 UTC. Signals notifications were delayed by up to 34 minutes. No alerts were lost. The incident was classified SEV-3.

59. INC-2026-03 delayed alerts - incident-03:1

chunk_id: incident-03:1
doc_id: incident-03
title: INC-2026-03 delayed alerts
category: incident
owner: Asterline Documentation
source_url: kb://asterline/incident/incident-03
published: 20260311
version: 1

A time synchronization configuration change moved scheduler wall clocks backward. The scheduler incorrectly skipped evaluation windows. Patch 2.3.1 introduced a monotonic clock and replay of missed windows.

60. INC-2026-03 delayed alerts - incident-03:2

chunk_id: incident-03:2
doc_id: incident-03
title: INC-2026-03 delayed alerts
category: incident
owner: Asterline Documentation
source_url: kb://asterline/incident/incident-03
published: 20260311
version: 1

Time synchronization configuration is now managed through reviewed infrastructure changes. The incident did not change the normal 4-minute evaluation interval or the 21-day seasonal anomaly history requirement.

61. Data retention policy - retention:0

chunk_id: retention:0
doc_id: retention
title: Data retention policy
category: policy
owner: Asterline Documentation
source_url: kb://asterline/policy/retention
published: 20260115
version: 2

Effective 2026-01-15, raw run records are held in cold recovery storage for 20 days after their plan-specific searchable retention period ends, then permanently deleted. This policy supersedes the earlier 60-day cold-storage period.

62. Data retention policy - retention:1

chunk_id: retention:1
doc_id: retention
title: Data retention policy
category: policy
owner: Asterline Documentation
source_url: kb://asterline/policy/retention
published: 20260115
version: 2

Deleted-account data is purged within 10 days. Backups are retained for 28 days. Cold recovery storage is not normally searchable through AQL, so it must not be added to the advertised searchable retention window.

63. Data retention policy - retention:2

chunk_id: retention:2
doc_id: retention
title: Data retention policy
category: policy
owner: Asterline Documentation
source_url: kb://asterline/policy/retention
published: 20260115
version: 2

These rules cover Asterline-managed storage. Copies exported to a customer warehouse are controlled by that customer. Run Explorer timeline retention is specified separately and is not extended by cold recovery storage.

64. Security overview - security:0

chunk_id: security:0
doc_id: security
title: Security overview
category: policy
owner: Asterline Documentation
source_url: kb://asterline/policy/security
published: 20250315
version: 1

Asterline encrypts data with AES-256 at rest and TLS 1.2 or later in transit. Project ingestion keys can be rotated by workspace admins. SAML SSO is available only on Enterprise.

65. Security overview - security:1

chunk_id: security:1
doc_id: security
title: Security overview
category: policy
owner: Asterline Documentation
source_url: kb://asterline/policy/security
published: 20250315
version: 1

This overview states that raw run records remain in cold recovery storage for 60 days after searchable retention ends. It was published on 2025-03-15 and has not been revised since publication.

66. Security overview - security:2

chunk_id: security:2
doc_id: security
title: Security overview
category: policy
owner: Asterline Documentation
source_url: kb://asterline/policy/security
published: 20250315
version: 1

Annual penetration testing is performed by Northbridge Security. Security reports should omit tokens, credentials, and customer payloads unless a secure transfer is arranged. This overview makes no claims about healthcare-specific certifications.

67. Employee handbook - handbook:0

chunk_id: handbook:0
doc_id: handbook
title: Employee handbook
category: policy
owner: Asterline Documentation
source_url: kb://asterline/policy/handbook
published: 20260105
version: 3

Employees receive 22 days of annual leave and 10 days of sick leave each year. Hybrid teams attend the office on Wednesdays and Thursdays. Managers approve exceptions based on team needs.

68. Employee handbook - handbook:1

chunk_id: handbook:1
doc_id: handbook
title: Employee handbook
category: policy
owner: Asterline Documentation
source_url: kb://asterline/policy/handbook
published: 20260105
version: 3

Effective 2026-01-01 the weekly on-call stipend is $175, replacing $140. This compensation update supersedes stipend figures in older operational runbooks. Incident severity definitions remain owned by the on-call runbook.

69. Employee handbook - handbook:2

chunk_id: handbook:2
doc_id: handbook
title: Employee handbook
category: policy
owner: Asterline Documentation
source_url: kb://asterline/policy/handbook
published: 20260105
version: 3

Primary parental leave is 24 weeks and secondary parental leave is 6 weeks. Employees should use the HR portal for personal leave requests rather than adding private medical details to shared operational notes.

70. Customer onboarding guide - onboarding:0

chunk_id: onboarding:0
doc_id: onboarding
title: Customer onboarding guide
category: onboarding
owner: Asterline Documentation
source_url: kb://asterline/onboarding/onboarding
published: 20260210
version: 2

Start by choosing a workspace plan and creating a project in the required supported region. Configure a project ingestion key through the application environment and send one test workflow run. Verify visibility before enabling full-volume ingestion.

71. Customer onboarding guide - onboarding:1

chunk_id: onboarding:1
doc_id: onboarding
title: Customer onboarding guide
category: onboarding
owner: Asterline Documentation
source_url: kb://asterline/onboarding/onboarding
published: 20260210
version: 2

Create a threshold Signal first; seasonal anomaly requires 21 complete days of history. For Team and Enterprise, configure Warehouse Export only after the customer grants the required warehouse permissions. Test a small export before enabling the regular schedule.

72. Customer onboarding guide - onboarding:2

chunk_id: onboarding:2
doc_id: onboarding
title: Customer onboarding guide
category: onboarding
owner: Asterline Documentation
source_url: kb://asterline/onboarding/onboarding
published: 20260210
version: 2

Check searchable record retention and Run Explorer retention separately. For a region move, create a new project and redirect future ingestion; historical records are not migrated automatically. The current pricing and region documents govern eligibility.

73. Product FAQ - faq:0

chunk_id: faq:0
doc_id: faq
title: Product FAQ
category: onboarding
owner: Asterline Documentation
source_url: kb://asterline/onboarding/faq
published: 20260311
version: 3

Asterline is a workflow-monitoring service. Run Explorer displays workflow attempts and status changes; it does not capture screens or DOM content. Signals notifications can be delayed during incidents even when the underlying run records remain available.

74. Product FAQ - faq:1

chunk_id: faq:1
doc_id: faq
title: Product FAQ
category: onboarding
owner: Asterline Documentation
source_url: kb://asterline/onboarding/faq
published: 20260311
version: 3

There is no announced timeline for an Asia-Pacific region. No mobile dashboard application is planned for 2026. Enterprise subscription prices must be obtained from sales and are not published in this knowledge base.

75. Product FAQ - faq:2

chunk_id: faq:2
doc_id: faq
title: Product FAQ
category: onboarding
owner: Asterline Documentation
source_url: kb://asterline/onboarding/faq
published: 20260311
version: 3

When documentation differs, compare the specific topic, effective date, scope, and explicit supersession statement. A later general overview does not automatically override a specific policy. Ask for clarification when the evidence cannot resolve a disagreement.